

PRISM-Loc: Three LiDAR Localization Backends Behind One ROS 2 Contract, with Middleware-Free Estimator Cores

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Abstract—Widely used open-source tools for map-based LiDAR localization of mobile robots are still largely organized by sensor suite: a 2D scanner typically calls for an AMCL-style particle filter over an occupancy grid, a 3D LiDAR for an NDT-based tracker over a point-cloud map, and an inertial- or GNSS-aided platform for a Kalman fusion node, often with different topics, parameters, initialization, and recovery behavior. Existing libraries already share one particle-filter core across 2D and 3D sensor models, and other systems already relocalize a robot by branch-and-bound scan matching. We describe `prism_loc`, an open-source ROS 2 localization stack that places three established estimator families behind one node contract: `laser2d`, a likelihood-field Monte Carlo localizer; `ndt3d`, a planar NDT-MCL backend [1] that reuses the same particle filter with a Gaussian-voxel measurement model; and `fusion3d`, a 15-state error-state Kalman filter fusing LiDAR, IMU, and RTK-GNSS. Every backend publishes the same `map→odom` transform and pose with covariance and accepts the same `/initialpose` seeding, so the sensor suite selects a backend at launch time rather than a stack with a different interface. All filter mathematics reside in middleware-free C++17 cores that depend only on Eigen and are exercised by deterministic, seeded unit and integration tests with no ROS runtime; the `laser2d` backend additionally recovers from the kidnapped-robot condition by deterministic branch-and-bound scan matching. We claim no new estimator theory; the contribution is this particular integration, released as usability-focused open-source software. No external baseline and no real or public data are used in the evaluation, and no field-accuracy claim is made.

I. Introduction

Localizing a mobile robot against a prior map is, on paper, a solved problem, but the widely used open-source tools are still largely organized by sensor suite. A team fielding a 2D LiDAR typically reaches for `nav2_amcl`, a particle filter that tracks a planar pose against an occupancy grid and nothing else. A team with a 3D LiDAR may reach instead for an `hdl_localization`-style tracker built around 3D NDT registration. A team that must fuse inertial and satellite measurements may reach for a `robot_localization`-style node [2], which composes an EKF over odometry, IMU, and GNSS but leaves the map-relative LiDAR measurement to another node. These tools differ in topic interface, parameter vocabulary, initialization, and recovery behavior, so a practitioner who moves from a 2D platform to a 3D one, or who adds a GNSS receiver, must re-integrate much of the localization layer.

The split is not universal, and several efforts reduce it. The survey of the ROS 2 navigation maintainers [3] catalogs the shortcomings of AMCL and reports a proposed design for a localization framework that builds 2D and 3D localization systems from modular components. `Beluga` [4], [5] is a ROS-agnostic C++17 Monte Carlo localization library whose ROS 2 nodes offer interface parity with `nav2_amcl` and pair one particle-filter core with likelihood-field and 2D and 3D NDT sensor models; Autoware’s pose-estimator arbiter [6] launches several heterogeneous pose estimators and provides a mechanism to stop and resume them; and `fuse` [7] and `MOLA` [8] provide modular estimation frameworks (Section II details these and the surrounding prior work). How a stack initializes, degrades, and recovers also matters in deployment: the Nav2 marathon deployment [9] names low localization confidence as one of the two causes behind most of its recovery triggers. Classical MCL recovers from such failures through stochastic mechanisms, random particle injection or dual sampling [10], [11], which are hard to reproduce in a test; later systems add failure detection [12], keep several particle populations seeded by map matching [13], or reset a tracker from a branch-and-bound global search [14], [15].

We describe `prism_loc`, an engineering integration that packages three map-based LiDAR localization backends behind one ROS 2 node contract: `laser2d`, a 2D likelihood-field Monte Carlo localizer against an occupancy grid; `ndt3d`, a 3D NDT-MCL backend [1] that reuses the same particle filter with a voxel-Gaussian measurement model over a prior point cloud, sampling the planar (x, y, θ) pose; and `fusion3d`, a 15-state error-state Kalman filter (ESKF) that fuses 3D LiDAR, IMU, and RTK-GNSS with IMU-bias estimation. All three present the same interface — a `map→odom` transform, a pose with 6×6 covariance on a private `~/pose` topic that differs between the two nodes only by namespace and can be remapped (Table II), and `/initialpose` seeding — so that choosing between a 2D scan, a 3D cloud, and an inertial/GNSS-aided configuration is a launch-time choice of parameter or launch file rather than a change of interface. The two MCL backends share a single pure-C++17 estimator core (motion model, particle filter, KLD-adaptive resampling, pose estimate), and the fusion backend is a second such core; in both cases the algorithmic code depends only on Eigen and is unit-

tested with no ROS or PCL present. When laser2d is started with no initial pose, it recovers one from a single scan by deterministic branch-and-bound (BBS) correlative scan matching over the occupancy grid, and re-seeds the particle filter from that result. We claim no new estimator theory: likelihood-field MCL, NDT-MCL, ESKF fusion, and BBS scan matching are all established, and so are the broader patterns we use, namely one particle-filter core shared by 2D and 3D measurement models [4], [5] and branch-and-bound relocalization against a prior map [16], [14]. We are not aware of a released package that combines exactly the three backends below behind one contract with a relocalizer that can decline to commit; that combination is the contribution, delivered as usable open-source software. Table I lists the closest prior work for each component.

Concretely, the paper describes:

- 1) A unified, pluggable localization architecture in which three estimator families (2D likelihood-field MCL, 3D NDT-MCL, and LiDAR+IMU+RTK-GNSS ESKF fusion) present one ROS 2 map→odom contract, so the sensor suite selects a backend, not a stack with a different interface. Beluga already serves its likelihood-field and NDT particle filters through one node interface [4], [5]; the addition here is an inertial/GNSS Kalman backend that shares the same map→odom TF edge, pose message type, and /initialpose seeding (laser2d and ndt3d are chosen by one backend parameter, fusion3d by its own launch file); because the consistency of its covariance has not been established, we release it as experimental (Section VII).
- 2) ROS-free estimator cores that hold all filter mathematics behind an Eigen-only, PCL-free, rclcpp-free boundary, making every algorithmic claim exercisable by deterministic, seeded-RNG unit and integration tests that run without a middleware runtime. Middleware-independent estimator cores are established practice [4], [17]; we list this as a property of the artifact, not as a research contribution.
- 3) A deterministic global relocalization path for laser2d: the branch-and-bound correlative scan matcher of Hess et al. [16], run over a max-pooled likelihood pyramid, recovers a pose for a kidnapped robot from one scan and re-seeds the MCL particle set with a repeatable search, in place of stochastic random-injection recovery. By the standard branch-and-bound argument of Hess et al. [16], the search returns the best correlative score over its discretized search window; this is the discrete optimum of a surrogate score, not an optimal pose estimate.
- 4) An open-source release engineered for usability, including fail-fast configuration validation, startup watchdog diagnostics that name the input that has gone silent, and a single-source parameter reference

kept next to the code that reads it.

Section II situates prism_loc against prior localization and fusion work. Section III describes the shared contract and the core/ROS boundary; Section IV details the estimator cores and Section V the ROS 2 nodes. Section VI presents the ROS-free testing and CI strategy that stands in for on-robot benchmarks in this software-description paper, together with a seeded synthetic evaluation that demonstrates closed-loop tracking, global relocalization, and fusion behavior of the actual cores, and Sections VII and VIII discuss limitations and conclude. The evaluation runs no external baseline and uses no real or public data, so we make no claim of accuracy or robustness relative to nav2_amcl, Beluga, MH-AMCL, or the global localization methods of Section II; that comparison is planned (Section VII).

II. Related Work

Monte Carlo localization. Particle-filter localization was introduced by Dellaert et al. [18] and Fox et al. [10], who represent the pose posterior as a weighted sample set updated by motion-model prediction, sensor-likelihood weighting, and resampling, and who demonstrated global localization from a uniform prior on real robots. Thrun et al. [11] later hardened MCL against accurate sensors and the kidnapped-robot problem via Mixture-MCL’s dual sampler. The particle-filter core in prism_loc is a direct descendant of this line of work and of its nav2_amcl lineage; we contribute no new filter theory. Sharing one such core across measurement models is not new either. Beluga [4], [5] is a ROS-agnostic, extensible C++17 MCL library; its ROS 2 nodes offer interface parity with nav2_amcl and combine one particle-filter core with likelihood-field, beam, and 2D and 3D NDT sensor models, and its 3D NDT node estimates a full $SE(3)$ pose, which is more general than the planar state of our ndt3d backend. Earlier, mcl_3dl [19] implemented 3D/6-DoF point-cloud MCL against a reference cloud for ROS, and Martín Rico et al. [20] replace Nav2’s AMCL with an MCL over Gridmaps and Octomaps for non-planar terrain. Other work makes MCL recovery more reliable: Akai [12] estimates the reliability of the MCL estimate to detect failure and integrates a global localization method through importance sampling for quick re-localization, and MH-AMCL [13] maintains several particle populations, created by multi-scale map matching, and outputs the most reliable one. prism_loc instead delegates recovery from localization failure to a deterministic branch-and-bound search (Section IV-B).

NDT representations and NDT-based measurement models. Biber and Straßer [21] introduced the Normal Distributions Transform, modeling a scan as per-cell Gaussians and reducing registration to Newton optimization of a smooth score; Magnusson et al. [22] generalized the representation to 3D voxel grids for field deployment on

mining vehicles. Both use NDT for deterministic scan-to-scan or scan-to-map registration that requires a reasonable initial guess. Saarinen et al. [1] brought the NDT into Monte Carlo localization (NDT-MCL), using it as the representation of both the map and the sensor data. `prism_loc`'s `ndt3d` backend follows this NDT-MCL idea in a simplified form: it scores raw subsampled points against the Gaussian-voxel map and reads the score out as a per-particle likelihood inside the shared MCL core, which trades the quadratic convergence of Newton-based alignment for the multi-hypothesis robustness of the particle filter, and lets the 2D and 3D backends differ only in their measurement model.

LiDAR-inertial and GNSS fusion. Solà's tutorial [23] is the mathematical basis of `prism_loc`'s fusion backend: the nominal/error-state decomposition, local angular-error parameterization, and predict-correct-inject-reset cycle are implemented essentially as derived there. On the systems side, Moore and Stouch's `robot_localization` [2] established the pattern of a generalized, configurable EKF/UKF fusion node for ROS, and Wan et al. [24] demonstrated an industrial ESKF fusing GNSS-RTK, LiDAR, and IMU at centimeter accuracy as a production system tied to its own pre-built LiDAR maps. Tightly-coupled LiDAR-inertial odometry such as FAST-LIO [25], and applied 3D map-based tracking systems such as Koide et al. [26], are complementary: they produce ego-motion and maps, whereas `prism_loc` localizes within a prior map. Relative to these, our `fusion3d` backend claims no estimator novelty; it packages a textbook ESKF as a ROS-free core behind the same ROS 2 contract as the two MCL backends, so that odometry-only, LiDAR-only, and GNSS-aided configurations are deployment choices rather than stacks with different interfaces.

Global (kidnapped-robot) localization. Olson [27] formulated 2D scan matching as exhaustive correlative search over a rasterized log-likelihood table with coarse-to-fine multi-resolution pruning, and Hess et al. [16] bounded that search with branch-and-bound over a precomputed grid pyramid to run loop closure in real time inside Cartographer. `prism_loc` borrows exactly this branch-and-bound-over-likelihood-pyramid primitive and applies it to relocalization against a known static map, a setting that existing tools already serve. `hdl_global_localization` [14] offers a 2D grid-map branch-and-bound engine, among others, as a ROS service, which `hdl_localization` [26] calls to reset the pose of its UKF-based tracker, whose corrections come from NDT scan matching; the global-localization tooling of `lidar_localization_ros2` [15], documented in 2026, runs map-wide branch-and-bound over an occupancy grid, returns ranked candidates, and re-seeds an NDT or GICP tracker through `/initialpose` behind safety guards that include consecutive confirmation samples and a cross-check against along-corridor aliases. 3D-BBS [28] extends branch-and-bound global localization to 3D point-cloud maps. Adurthi [29] uses a hierarchical,

max-pooled scan-to-map matcher to enrich the particles of a 3D-LiDAR particle filter before resampling. Other single-scan 2D methods include CBGL [30] and the method of Zhang et al. [31], which samples positional hypotheses over traversable space with a rapidly-exploring random tree. Feeding global-matching or global-localization results into a particle filter is itself established [29], [13], [12]; what `prism_loc` adds is a specific wiring to its own MCL core: a 2D branch-and-bound search, with no submaps or pose graph to maintain, whose accepted result re-seeds the particle set once, in place of the heuristic recovery mechanisms of classical MCL [10], [11].

Multi-hypothesis verification and abstention. Deferring commitment until later observations disambiguate a place is also established. Gao et al. [32] extract candidate poses from successive 2D laser scans and resolve them with an extended multiple-hypothesis tracker whose likelihood combines consecutive candidate poses with odometry; MH-AMCL [13] keeps several particle populations alive and selects the most reliable; Adurthi [29] declares a vehicle localized only after the particle spread stays small over consecutive scans and the vehicle has moved; Reliable-loc [33] monitors sequential pose uncertainty to switch localization modes in large street scenes; and Akai [12] estimates the reliability of an MCL estimate to detect failure. Concurrently with this work, Zhang et al. [34] proposed a selective layer for LiDAR-to-BIM global initialization that aggregates candidates from several submaps and issues typed COMMIT, DEFER, or AMBIGUOUS decisions, which symmetry-breaking motion can upgrade to COMMIT. The relocalization decision described in Section IV-B is simpler than all of these: the best match of a single scan is accepted or rejected by a score threshold, and no hypotheses are tracked over later scans.

Robot software systems. Nav2 [9] treats localization as an external, swappable component (stock AMCL or SLAM Toolbox), and its marathon deployment names low localization confidence as one of the two causes behind most of its recovery triggers. The ROS 2 maintainers' survey [3] catalogs AMCL's shortcomings and reports a proposed design for a localization framework that builds 2D and 3D localization systems atop modular components. Several systems already place more than one estimator behind a common interface. `fuse` [7] is a ROS framework for sensor fusion by nonlinear least squares with plugin-based sensor and motion models; Autoware's pose-estimator arbiter [6] launches NDT scan matching, YabLoc, EagleEye, and a landmark-based localizer, selected by a `pose_source` argument, and provides a stop/resume mechanism driven by pluggable switching rules (the shipped rule enables all of them); and MOLA [8] is a framework for LiDAR odometry, map manipulation, and localization whose pipelines are assembled from reusable blocks. Estimator cores kept free of the middleware are likewise established practice [4], [17]. As framed in Section I, `prism_loc` is one narrow instance of this pattern rather than a new algorithm:

it places exactly three estimator families (likelihood-field MCL, NDT-MCL, ESKF fusion) behind one ROS 2 topic and TF contract, with each estimator implemented as a ROS-free library that is unit-testable without a middleware runtime. It covers a small part of the direction set out by the modular 2D/3D localization framework whose design the survey [3] reports as proposed, and it is not a general localization framework. The accuracy evaluation planned in Section VII will follow the trajectory-error conventions (ATE/RPE) standardized by the TUM RGB-D benchmark [35], applied to per-backend map→odom estimates rather than visual SLAM output.

III. System Architecture

`prism_loc` is organized as two layers realized by four packages (Fig. 1). The lower layer holds the estimator cores: `prism_loc_core`, which implements the Monte Carlo localization (MCL) family, and `prism_loc_fusion`, which implements the error-state Kalman filter (ESKF). Both are plain C++17 libraries whose only third-party dependency is Eigen; each CMakeLists.txt calls `find_package(Eigen3 REQUIRED)` and links `Eigen3::Eigen` alone, with no reference to `roscpp`, `tf2`, or `PCL`. The upper layer holds the ROS 2 node packages: `prism_loc` (the `laser2d` and `ndt3d` backends) and `prism_loc_fusion_ros` (the `fusion3d` backend). These packages, and only these, depend on `roscpp`, the `tf2` stack, the ROS message packages, and `PCL`. Each core is consumed by its node package through the ordinary `ament find_package` mechanism, so the dependency direction is strictly upward: the cores never link against the middleware.

Why the split. Isolating the estimators from the middleware buys three properties. First, unit-testability without a running graph: each core builds and tests with a plain system toolchain and `gtest`, so its numerical behavior is exercised with no ROS installation, no node, and no message plumbing. The CMakeLists.txt of each core wires tests through plain `CTest` with system `gtest`, enabled by default via `BUILD_TESTING`, and runnable both inside a `colcon` workspace and standalone with a bare `cmake+ctest` invocation; continuous integration exercises this ROS-free path directly. The detailed testing story is deferred to Section VI. Second, determinism: all sampling routes through a single seeded `std::mt19937_64` wrapper (default seed 42), so a motion-prediction, weighting, or resampling step is reproducible bit for bit across runs, which is what makes the core tests assertable rather than merely statistical. Third, one output contract shared by all three backends. Regardless of backend, a node publishes the map→odom transform on `/tf`, a `PoseWithCovarianceStamped` on `~/pose`, and accepts a `PoseWithCovarianceStamped` on `/initialpose` for seeding; the MCL nodes additionally publish a particle cloud while the fusion node publishes an `Odometry` message carrying pose and velocity. Because the contract is fixed, choosing between the

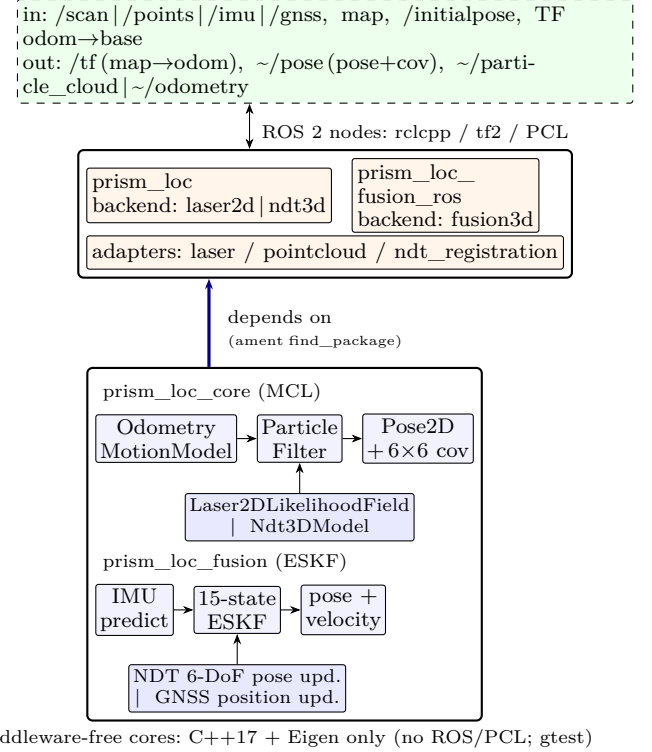


Fig. 1. `prism_loc` architecture. Two middleware-free estimator cores (bottom, C++17 + Eigen only): `prism_loc_core` runs the MCL loop (`OdometryMotionModel`→`ParticleFilter`, corrected by either the `Laser2DLikelihoodField` or `Ndt3DModel` measurement model, plus a branch-and-bound relocalizer), and `prism_loc_fusion` runs a 15-state ESKF (IMU predict, corrected by NDT 6-DoF pose and GNSS position updates). The ROS 2 node packages (top) consume the cores upward through `ament`, select a backend at runtime, bridge messages via adapters, and expose one shared contract: a map→odom TF, a pose-with-covariance output, and an `/initialpose` seed input.

likelihood-field MCL, NDT-MCL, and ESKF estimators is a launch-time choice: `prism_loc` reads a backend string parameter ("`laser2d`" by default) and branches on it when constructing its measurement model and when applying the correction step, and `prism_loc_fusion_ros`, started by its own launch file, supplies the `fusion3d` estimator behind the same topic and TF interface, up to the node namespace under which `~/pose` resolves (Table II). As stated in Section I, this factoring — three known estimator families behind one ROS 2 contract — is a specific integration of established patterns rather than a new architecture (Table I); the estimators themselves are textbook.

Core interfaces. The MCL core expresses the shared predict–weight–resample loop against a small set of value types. A `Pose2D` carries the planar pose (x, y, θ) ; a `Particle` pairs a pose with a weight; and the two backends differ only through a `MeasurementModel` abstraction whose single virtual method returns the log-likelihood of an observation given a candidate pose. `Laser2DLikelihoodField` (a 2D scan against an occupancy-grid likelihood field) and `Ndt3DModel` (a 3D cloud against a Gaussian-voxel

TABLE I

Closest prior work for each component of prism_loc. The last column states what prism_loc does, not what the cited systems lack; a component is claimed as a difference only where the cited sources document the alternative.

Component	Closest prior work	In prism_loc
One particle-filter core for 2D likelihood-field and NDT models	Beluga [4]; NDT-MCL [1]	No methodological difference. ndt3d samples a planar pose (Beluga’s 3D NDT node estimates $SE(3)$ [5]) and scores raw points against the NDT map.
Several estimators behind one interface	fuse [7]; Autoware pose-estimator arbiter [6]; MOLA [8]	Exactly three backends, 2D MCL, planar NDT-MCL, and a LiDAR+IMU+RTK-GNSS ESKF, behind one map→odom contract chosen at launch.
Middleware-free estimator core	Beluga [4]; iris_lama [17]	No methodological difference; one seeded generator makes the tests and experiments bit-reproducible.
Branch-and-bound relocalization against a prior map	Hess et al. [16]; hdl_global_localization [14]; lidar_localization_ros2 [15]; 3D-BBS [28]; Adurthi [29]; MH-AMCL [13]; Akai [12]	Lattice-exact 2D branch-and-bound over the configured search window, whose accepted result re-seeds the shared MCL core once; feeding global matches into a particle filter is itself established.
Error-state Kalman fusion of LiDAR, IMU, and GNSS	Solà [23]; Wan et al. [24]	No methodological difference; the filter is exposed under the same contract as the MCL backends and is released as experimental, because the consistency of its covariance has not been established (Section VII).

NDT map) are its two implementations, and the same ParticleFilter and OdometryMotionModel drive both. The fusion core exposes an ESKF class over a 15-dimensional error state with three entry points — an IMU-driven predict, a 6-DoF updatePose fed by NDT registration, and a 3-DoF updatePosition fed by an absolute position fix — alongside a geodetic helper that converts WGS84 latitude/longitude/ altitude to a local ENU frame. Section IV details these estimators.

Adapters. The node layer bridges ROS message types to the middleware-free core types through thin adapters, keeping the cores ignorant of message definitions. In prism_loc, a laser adapter converts a LaserScan to the core’s LaserScan2D (honoring range overrides) and an OccupancyGrid to the core grid map, and a pointcloud adapter converts a PointCloud2, and a .pcd map file, into the std::vector<Eigen::Vector3d> the NDT map ingests; both also convert quaternion-bearing Transform and Pose messages to Pose2D by extracting yaw. In prism_loc_fusion_ros, an ndt_registration adapter wraps PCL’s NDT to align an incoming cloud against the prior-map cloud and returns the resulting Isometry3d pose, which becomes the ESKF’s 6-DoF measurement. PCL therefore appears only inside these adapters, never in the cores. The node-layer responsibilities — TF lookups, topic wiring, and runtime diagnostics — are described in Section V.

IV. Estimator Cores

This section describes the three estimator families that prism_loc places behind a single ROS 2 contract. All three are implemented as ROS-free C++ libraries: the two Monte Carlo localization (MCL) backends and the global relocalizer live in prism_loc_core, and the fusion filter lives in prism_loc_fusion. Consistent with the architectural framing of Section III, the contribution here is an

integration of established components, not new estimation theory: we reuse textbook constructions and report only what the code implements.

A. Shared Monte Carlo Localization Core

The two 2D and 3D pose-tracking backends share one particle filter. Following Thrun-style MCL, the pose posterior at time t is a weighted sample set $\{(x_t^{[i]}, w_t^{[i]})\}_{i=1}^N$ over the planar pose $x_t = (x, y, \theta)$ (Pose2D), updated by the standard predict–weight–resample recursion. Only the measurement model differs between backends, so the entire filter is written against an abstract interface rather than any specific sensor.

a) Motion model: OdometryMotionModel implements the sampling-based odometry model. Each control u_t is decomposed into a first rotation, a translation, and a second rotation $(\delta_{r1}, \delta_t, \delta_{r2})$ from consecutive odometry poses, and each particle is advanced by the noise-corrupted increment

$$\begin{aligned}\hat{\delta}_{r1} &= \delta_{r1} - \varepsilon(\alpha_1 \delta_{r1}^2 + \alpha_2 \delta_t^2), \\ \hat{\delta}_t &= \delta_t - \varepsilon(\alpha_3 \delta_t^2 + \alpha_4 (\delta_{r1}^2 + \delta_{r2}^2)), \\ \hat{\delta}_{r2} &= \delta_{r2} - \varepsilon(\alpha_1 \delta_{r2}^2 + \alpha_2 \delta_t^2),\end{aligned}\tag{1}$$

where $\varepsilon(\sigma^2)$ draws zero-mean Gaussian noise with the given variance and α_{1-4} (MotionParams) are the usual rotation/translation noise coefficients. A pure-rotation guard sets $\delta_{r1} = 0$ when the translation is below 10^{-3} m to avoid an ill-defined heading.

b) Particle filter: ParticleFilter weights each particle by the measurement model’s log-likelihood, subtracting the per-step maximum before exponentiation for numerical stability, then normalizing. Resampling is triggered only when the effective sample size $N_{\text{eff}} = (\sum_i w^{[i]})^2 / \sum_i (w^{[i]})^2$ drops below resample_threshold

times N , and is KLD-adaptive [36]: samples are drawn proportional to weight until the count reaches the Kullback–Leibler bound

$$M_x = \frac{k-1}{2 \varepsilon_{\text{kld}}} \left(1 - \frac{2}{9(k-1)} + \sqrt{\frac{2}{9(k-1)}} z_{1-\delta} \right)^3, \quad (2)$$

where k is the number of occupied histogram bins seen so far. Bins are formed by discretizing (x, y) at `kld_bin_xy` and θ at `kld_bin_yaw`; the sample count is clamped to `[min_particles, max_particles]`, and $z_{1-\delta}$ (`kld_z`) and ε_{kld} (`kld_err`) are the standard-normal quantile and error bound. The pose estimate is the weighted mean of (x, y) and the circular mean of θ ; the reported 6×6 covariance carries the planar terms and marks z , roll, and pitch as unobserved.

c) Measurement models: A backend supplies a `MeasurementModel`, whose sole method returns the unnormalized log-likelihood of the current observation given a candidate pose. Two are provided. `Laser2DLikelihoodField` is a Thrun-style likelihood-field model for 2D scans [11]: an occupancy grid is preprocessed into a `LikelihoodField` that stores, per cell, the Euclidean distance d to the nearest occupied cell (computed once by a separable distance transform), and each subsampled beam endpoint contributes $\log(z_{\text{hit}} e^{-d^2/2\sigma_{\text{hit}}^2} + z_{\text{rand}}/z_{\text{max}})$, with `z_hit`, `z_rand`, `sigma_hit`, and `max_beams` as parameters. `Ndt3DModel` reuses the identical filter but scores a 3D point cloud against a Normal Distributions Transform map [21], [22]. The prior map is voxelized into per-voxel Gaussians (μ_v, Σ_v) (`NdtMap`, with small eigenvalues inflated to a fraction of the largest for numerical conditioning), and the per-particle likelihood accumulates $\log(s(p) + s_{\text{floor}})$ over cloud points p , where $s(p) = \exp(-\frac{1}{2}(p - \mu_v)^\top \Sigma_v^{-1}(p - \mu_v))$ and empty voxels score zero. The planar particle state (x, y, θ) is lifted to a 3D transform at a fixed base height, so the 3D backend estimates the same (x, y, θ) as the 2D one. This is a simplified form of NDT-MCL [1], which represents both the map and the scan by NDTs; here raw points are scored against the NDT map, and a score of the kind that Newton registration optimizes is read out as a particle likelihood, so the two backends differ only in this class.

B. Branch-and-Bound Global Localization

Recovery from a lost or kidnapped estimate is handled by a deterministic correlative scan matcher, `BranchAndBoundMatcher`, following Olson’s multi-resolution correlative search [27] and the branch-and-bound formulation of Hess et al. [16]. At construction the likelihood field is converted into a probability grid $p = e^{-d^2/2\sigma_{\text{hit}}^2}$, and a max-pooling pyramid of `max_depth+1` levels is precomputed, where level h stores the maximum over a $2^h \times 2^h$ cell block. A search then proceeds over discretized poses: translations within `linear_window` and rotations spanning `angular_window` at `angular_step` resolution. Coarse nodes

are expanded depth-first with children visited in decreasing order of their pyramid-derived upper bound; because the max-pooled bound is admissible, a subtree is pruned only when its bound cannot exceed the best exact (leaf) score found so far, so, by the standard branch-and-bound argument of Hess et al. [16], the returned pose attains the best correlative score over the discretized search window. This is the discrete optimum of a correlative surrogate score, not of the pose posterior; it is not optimal localization in any statistical sense. The result is accepted only when its score reaches `min_score_fraction` of the number of beams used; an accepted pose re-seeds the particle filter, whereas a rejected one leaves the current estimate untouched. Unlike the stochastic random-injection recovery of classical MCL, this search is deterministic and therefore directly reproducible in a unit test.

C. Error-State Kalman Filter Fusion

The fusion backend, `Eskf`, is a 15-state error-state Kalman filter that follows Solà’s quaternion formulation [23]. The nominal state (`NominalState`) carries position p , velocity v , orientation quaternion q , accelerometer bias b_a , and gyroscope bias b_g ; the error state $\delta x \in \mathbb{R}^{15}$ collects $(\delta p, \delta v, \delta \theta, \delta b_a, \delta b_g)$ with a local angular error $\delta \theta$. On each IMU sample the nominal state is integrated in closed form,

$$p \leftarrow p + v \Delta t + \frac{1}{2} a_w \Delta t^2, \quad v \leftarrow v + a_w \Delta t, \quad q \leftarrow q \otimes \text{Exp}(\omega \Delta t), \quad (3)$$

where $a_w = R(q)(a_m - b_a) + g$ and $\omega = \omega_m - b_g$ are the bias-corrected, gravity-compensated measurements, and $\text{Exp}(\cdot)$ is the $SO(3)$ exponential (`so3.hpp`). The error covariance is propagated with the discrete transition F and additive process noise Q built from the IMU noise parameters,

$$P \leftarrow F P F^\top + Q. \quad (4)$$

Two measurement updates share the standard Kalman gain $K = P H^\top (H P H^\top + R)^{-1}$, $\delta x = K y$, $P \leftarrow (I - K H) P$: `updatePose` applies a 6-DoF NDT pose observation whose residual stacks the position error and the local orientation error $\log(q^{-1} q_{\text{meas}})$, while `updatePosition` applies a 3-DoF GNSS position observation. GNSS fixes are mapped from geodetic coordinates into a local ENU frame through a WGS-84 LLA→ECEF→ENU conversion about a fixed datum (`GeodeticConverter`). After each update the estimated error is injected into the nominal state — additively for the vector components and multiplicatively, $q \leftarrow q \otimes \text{Exp}(\delta \theta)$, for the orientation — and the error state is reset to zero. Because the filter takes plain vectors and quaternions rather than ROS messages, the entire predict–correct–inject–reset cycle is exercised in ROS-free unit tests.

V. ROS 2 Integration and Interface Contract

The estimator cores of Section IV carry no middleware dependency; they are made deployable by thin ROS 2 node packages (`prism_loc` for the laser2d and `ndt3d`

MCL backends, `prism_loc_fusion_ros` for the fusion3d ESKF backend) whose job is to translate messages, TF, and parameters into core calls and back. We treat this boundary as a first-class interface rather than glue code, following the conventions that `robot_localization` [2] and `nav2_amcl` within Nav2 [9] established for drop-in localization: a REP-105 frame contract, a `map→odom` TF output, and `/initialpose` seeding. A robot already configured for AMCL can therefore switch backends by changing a launch file, not its TF tree; consumers of the pose topic need at most a remapping, because `~/pose` resolves under the node name (Table II).

A. Frames and `map→odom` composition

All three backends honor the REP-105 frame chain `map→odom→base_link`, exposed as the `global_frame`, `odom_frame`, and `base_frame` parameters. Each backend estimates the robot body in the map, $T_{\text{map},\text{base}}$, but—so as not to fight the odometry source that owns `odom→base_link`—it broadcasts only the correction transform $T_{\text{map},\text{odom}}$. Writing $\oplus$ for pose composition and $(\cdot)^{-1}$ for inversion, the node computes

$$T_{\text{map},\text{odom}} = T_{\text{map},\text{base}} \oplus T_{\text{odom},\text{base}}^{-1}, \quad (5)$$

where $T_{\text{odom},\text{base}}$ is read from TF at the measurement stamp by both nodes; the fusion node, which publishes at IMU rate, falls back to the latest buffered transform—without blocking—when the odometry publisher has not yet caught up to the stamp. Equation (5) guarantees the round-trip identity $T_{\text{map},\text{odom}} \oplus T_{\text{odom},\text{base}} = T_{\text{map},\text{base}}$, so downstream consumers recover the intended body pose; both the planar core (compose/inverse on $SE(2)$) and the fusion node (`map_base-odom_base-1` on $SE(3)$ Isometries) implement it, and each is checked by a ROS-free unit test asserting the round trip to 10^{-9} . If `odom→base_link` is momentarily unavailable, the fusion node degrades gracefully to publishing `map→base_link` directly rather than emitting a wrong transform. The broadcast stamp is future-dated by `transform_tolerance` (default 0.1 s) to bound extrapolation on the consumer side.

B. Topic contract and QoS

Table II lists the per-backend contract. Inputs and outputs are uniform across backends wherever the sensing allows: every backend outputs the same `map→odom` TF and a `PoseWithCovarianceStamped` on `~/pose`, a private topic that resolves under the node name, and every backend accepts `/initialpose`. QoS profiles match each publisher’s expected semantics rather than using defaults: sensor streams (`LaserScan`, `PointCloud2`, `Imu`, `NavSatFix`) are subscribed with `SensorDataQoS` (best-effort, volatile), while the occupancy-grid map is subscribed `transient_local` and `reliable` so that the single latched grid published once by `map_server` is delivered even though the subscriber joins later. A mismatch here is a common and silent failure mode for new ROS users; fixing the profiles at the interface removes it.

TABLE II

Per-backend topic contract. All backends additionally subscribe `/initialpose` and the TF `odom→base_link`, and broadcast `map→odom`. The private topic `~/pose` resolves under the node name (`/prism_loc/pose` for `laser2d` and `ndt3d`, `/prism_loc_fusion/pose` for `fusion3d`) and can be remapped.

Backend	Inputs	Outputs
<code>laser2d</code>	<code>/scan</code> , <code>/map</code> (<code>OccupancyGrid</code>)	<code>~/pose</code> , <code>~/particle_cloud</code>
<code>ndt3d</code>	<code>/points</code> , <code>.pcd</code> (file)	<code>~/pose</code> , <code>~/particle_cloud</code>
<code>fusion3d</code>	<code>/points</code> , <code>/imu</code> , <code>/gnss</code> (<code>NavSatFix</code>), <code>.pcd</code>	<code>~/pose</code> , <code>~/odometry</code>

C. Initialization paths

The interface offers one manual and one automatic seeding path per backend. Manually, all backends accept a `PoseWithCovarianceStamped` on `/initialpose` (the RViz “2D Pose Estimate” tool); the MCL backends spread a Gaussian particle cloud at the supplied mean and covariance, and the fusion backend loads the pose as its initial nominal state. Automatically, the two backends differ because their sensing differs. The `laser2d` backend can self-recover with no prior pose by running the branch-and-bound search of Section IV-B (try `global_localization`, or the on-demand `~/global_localization` service), which re-seeds the filter from a single scan. The service flags a request that is served on the next scan; when `try_global_localization` is false (the default) the matcher is constructed on demand at the first service call, so the service works regardless of the parameter, which then only controls automatic relocalization before the first pose. The `fusion3d` backend auto-initializes once it has both an attitude estimate—roll and pitch recovered from the IMU specific-force vector, heading from `initial_yaw`—and a position fix from the first accepted GNSS message, converted from WGS-84 to a local ENU frame; `/initialpose` overrides this when GNSS is unavailable.

D. Parameter surface

Every tunable is a declared ROS 2 parameter with an in-code default, documented in a single source of truth (`PARAMS.md`) maintained one-to-one against the `declare_parameter` calls. The surface deliberately exposes the core’s control knobs rather than hiding them: the KLD-sampling bounds (`kld_err`, `kld_z`, and the histogram bin sizes), the likelihood-field and laser-range parameters (`z_hit`, `sigma_hit`, `laser_min_range`, `laser_max_range`), the branch-and-bound search window and depth (`bbs_linear_window`, `bbs_angular_step`, `bbs_max_depth`, `bbs_min_score_fraction`), and the ESKF process noise and GNSS gates (`sigma_acc`, `gnss_min_status`, `gnss_max_pos_cov`). Because the cores are configured purely through plain structs, each parameter maps to exactly one field, and the shipped

YAML files override only the subset a deployment typically changes.

E. Runtime diagnostics

A localization node that silently does nothing is the dominant first-run frustration; this release invests in making every stalled precondition observable and self-explaining. Four mechanisms cooperate. (i) Fail-fast on a missing mandatory map: the `ndt3d` and `fusion3d` nodes abort construction with an actionable `std::runtime_error` if `map_pcd_path` is empty or fails to load, rather than idling forever. (ii) Throttled waiting-state warnings: whenever a callback cannot yet act—no map, no initial pose, no filter initialization—it emits a rate-limited WARN that names both the unmet precondition and the remedy (e.g. “waiting for an initial pose—publish to `/initialpose` ...or enable `try_global_localization`”). (iii) A 10 s startup watchdog reports, for each mandatory input that has produced no message, the topic name and its live publisher count (via `count_publishers`), disambiguating “no publisher” from “publisher present but QoS-incompatible”; it cancels itself once all inputs are seen. (iv) Quantified rejection logging: the GNSS gate logs each rejected fix with the measured value against the threshold it violated—status below `gnss_min_status`, NaN coordinates, or position variance above `gnss_max_pos_cov`—and the NDT update likewise reports non-convergence and fitness versus `ndt_max_fitness`. These diagnostics carry no algorithmic weight; they exist so that a misconfiguration is diagnosable from the console log alone, in the spirit of the logging infrastructure Nav2 relied on in its long-duration field experiment [9].

VI. Validation: Correctness Testing and Synthetic Evaluation

`prism_loc` is validated at the level of correctness, not field accuracy, in two complementary ways. First, a deterministic unit- and integration-test suite pins each estimator component to its specified behavior (Section VI-A). Second, a set of closed-loop synthetic experiments drives the same estimator cores through end-to-end localization scenarios and reports quantitative behavior (Section VI-B). Both run with no middleware, no simulator, and no recorded bags, because every estimator is a ROS-free library (Section IV).

A. Correctness Testing

The test suite comprises 44 GoogleTest cases across four packages: 25 in the shared `prism_loc_core` (the two MCL backends and global recovery), 13 in `prism_loc_fusion` (the ESKF), 4 in the `prism_loc` ROS node’s adapter layer, and 2 in `prism_loc_fusion_ros`.

What the cores are pinned to. The core tests target the components most likely to regress silently. The odometry motion model is checked both deterministically (zero-command and pure-translation cases) and statistically

(the mean of noisy samples approximates the commanded motion). Measurement models are checked for the monotonicity property they must satisfy to be usable as a likelihood: the ground-truth pose must out-score a displaced one, for both the 2D likelihood field and the 3D NDT voxel score. The particle-filter mechanics—low-variance resampling concentrating on a heavy particle, and the weighted-mean estimate—are tested directly, as are the occupancy-grid and NDT-map primitives (world/map round-trips, out-of-bounds handling, likelihood-field distances, and voxel-mean recovery). Global recovery is exercised by having branch-and-bound recover the correct global pose from a single scan—the kidnapped-robot scenario—and reject an empty scan. Two integration tests close the loop end to end, driving the full predict-weight-resample cycle until the estimate converges to ground truth for the `laser2d` and `ndt3d` backends respectively.

Fusion and math. The ESKF is pinned for both convergence and consistency: a static IMU leaves the state at rest, constant acceleration reproduces the expected kinematics, and position and pose updates pull the state toward the measurement; an integration test tracks a moving target from a deliberately wrong initial guess. Supporting identities are verified independently— $SO(3)$ exponential/logarithm round-trips and the skew/cross-product relation, geodetic ENU round-trips against a datum, and the map→odom TF composition arithmetic used to publish the correction transform.

Determinism. Every stochastic path draws from a single seedable generator exposed as the ROS parameter `seed` (default 42). The RNG tests assert that equal seeds reproduce identical sequences and that distinct seeds differ, so the statistical assertions above are reproducible run to run rather than flaky.

Continuous integration. Two CI jobs guard the contract from both sides. A core job builds `prism_loc_core` and `prism_loc_fusion` with plain CMake and runs `cctest`, requiring only a compiler and Eigen—no ROS—which keeps the cores honestly middleware-independent. A separate ROS job builds inside a `ros:humble` container, resolves dependencies with `rosdep`, and runs `colcon build` and `colcon test` across all four packages, so the ROS 2 wrapping is tested against a real distribution.

B. Synthetic Evaluation

Beyond per-component assertions, three closed-loop experiments drive the actual estimator cores—compiled directly from `prism_loc_core` and `prism_loc_fusion` sources with `g++` and Eigen, no ROS and no CMake—through end-to-end localization scenarios. All data are synthetic: worlds, scans, and inertial streams are generated by the experiment drivers themselves, with all randomness derived from fixed compile-time seeds, so every number and figure below is exactly reproducible from the drivers and CSV logs shipped in the repository (experiments/). We stress that this is a correctness and behavior demon-

stration on synthetic data, not a field-accuracy benchmark: no real robot or sensor is involved, and the sensor-noise models match the assumptions of the filters by construction. No external baseline is run: neither a tracker such as `nav2_amcl`, Beluga [4], MH-AMCL [13], or `hdl_localization` [26], nor a global localizer such as `hdl_global_localization` [14] or CBGL [30]. Every result below concerns `prism_loc`’s own cores, and a comparative study on real and public data is left to future work (Section VII).

MCL pose tracking and KLD adaptation. The first experiment exercises the `laser2d` pipeline exactly as `localization_node.cpp` wires it. A $12\text{ m} \times 10\text{ m}$ occupancy grid (0.1 m/cell, the integration test’s room-border convention plus two interior pillars that break pose symmetry) hosts a simulated robot driven around a closed oval loop with semi-axes 3.6 m and 2.6 m ($7.2\text{ m} \times 5.2\text{ m}$ extent) for 65 s at 10 Hz (650 steps). At each step a 180-beam scan is ray-cast at the true pose, and a drifting odometry frame is synthesized by integrating noise-corrupted motion increments under the `laser2d.yaml` odometry model ($\alpha_{1..4} = 0.2$). The real ParticleFilter with the Laser2DLikelihoodField model and KLD-adaptive resampling—every parameter taken from `laser2d.yaml`—is seeded with a broad Gaussian prior ($\sigma = 1\text{ m}$, 0.4 rad) centered at the true start pose and updated in the node’s predict–correct–resample order, gated by `update_min_d` = 0.2 m and `update_min_a` = 0.2 rad. Three fixed seeds (101, 202, 303) are run. Fig. 2 shows the resulting position and yaw error: after convergence ($t \geq 10\text{ s}$, across all three seeds) the mean position error is 0.105 m (max 0.29 m) and the mean yaw error is 0.032 rad = 1.8° (max 0.18 rad). Fig. 3 shows the KLD-adaptive particle count over the same runs: the set starts at the 2000-particle ceiling under the broad prior, collapses to the 500-particle floor after the first post-convergence update, and holds there for the rest of the loop—a $4\times$ reduction in per-update likelihood evaluations once the posterior is concentrated.

Global relocation by branch-and-bound. The second experiment evaluates the BBS matcher (Section IV-B) as a kidnapped-robot relocalizer. On the asymmetric $6\text{ m} \times 6\text{ m}$ room from the BBS unit test (two interior wall stubs break the four-fold symmetry, so a single 360° scan pins the global pose uniquely), $N = 100$ true poses are drawn uniformly over free space with uniform yaw (fixed seed 20240607), and one 180-beam scan is ray-cast per pose. To span an easy-to-hard range, each scan is corrupted with synthetic clutter: a per-query clutter fraction drawn uniformly in $[0, 0.8]$ replaces each beam, with that probability, by a uniform-random spurious range—modeling people, glass, and objects absent from the static map. The real BranchAndBoundMatcher then searches the whole map with the `laser2d.yaml` `bbs_*` defaults. Defining success as position error $< 0.5\text{ m}$ and yaw error $< 10^\circ$, 97 of 100 queries succeed (97.0%); over all 100 queries the recovered pose has median position error 0.037 m (95th

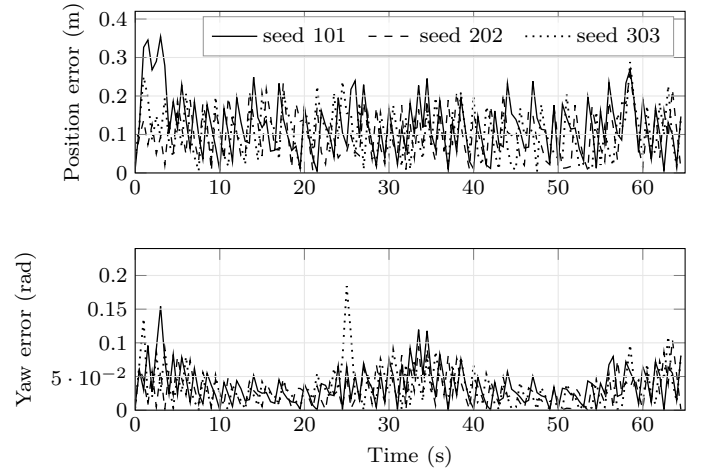


Fig. 2. Monte-Carlo-localization (MCL) tracking accuracy of the `laser2d` core on a synthetic occupancy grid. A simulated robot is driven around a closed oval loop with semi-axes 3.6 m and 2.6 m ($7.2\text{ m} \times 5.2\text{ m}$ extent) for 65 s at 10 Hz through a $12\text{ m} \times 10\text{ m}$ room (two interior pillars break pose symmetry); 180-beam scans are generated by grid-march ray casting (`test_helpers.hpp`), and the odometry is a drifting frame produced by integrating noise-corrupted motion increments under the `laser2d.yaml` α model. Every plotted value comes from the CSV written by `experiments/mcl_tracking.cpp`, which runs the real `prism_loc_core::ParticleFilter` with the Laser2DLikelihoodField model and KLD resampling, wired (predict $\rightarrow$ correct $\rightarrow$ resample, gated by `update_min_d`/`update_min_a`) exactly as `localization_node.cpp`. Three fixed seeds are shown as thin lines. Starting from a broad 1 m prior the filter converges within a few updates and then tracks the ground truth for the whole loop: across the three seeds, steady-state ($t \geq 10\text{ s}$) mean position error is 0.105 m (max 0.29 m) and mean yaw error is 0.032 rad (1.8° , max 0.18 rad).

percentile 0.072 m) and median yaw error 0.48° (95th percentile 1.74°). Fig. 4 plots position error against the BBS score fraction (raw score over used beams): the acceptance threshold `bbs_min_score_fraction` = 0.40 yields zero false accepts—all 3 failures score in 0.21–0.27 with $> 3\text{ m}$ and roughly 90° – 180° error, while 78 matches are accepted (all correct) and 19 correct fixes are conservatively rejected. The median score fraction is 0.632; successes span 0.22–0.97 while all failures fall below 0.28. Every query scoring at or above the 0.40 acceptance threshold is correct, so the threshold separates accepted matches from failures with a wide margin, even though three low-scoring successes overlap the failure score band. Median full-map query latency is 37.65 ms (min 8.77 ms, 95th percentile 106.06 ms; single-threaded, machine-dependent wall-clock; all accuracy numbers are deterministic across runs).

ESKF fusion under aiding dropout. The third experiment drives the actual `prism_loc_fusion` ESKF over a synthetic 60 s 3-D trajectory: a horizontal figure-eight ($x = 8\sin\omega t$, $y = 4\sin 2\omega t$, $\omega = 2\pi/30$) with a gentle $\pm 1.5\text{ m}$ altitude oscillation. The ideal body-frame specific force and angular rate computed from this analytic trajectory are corrupted with exactly the white-noise

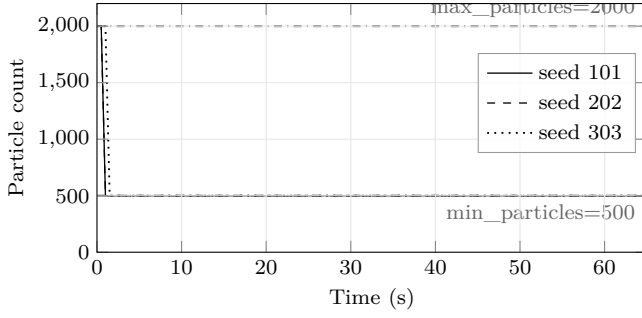


Fig. 3. KLD-adaptive resampling sizes the particle set to the current uncertainty (synthetic data; same run as Fig. 2, values from the CSV written by `experiments/mcl_tracking.cpp`). The dash-dotted lines mark the `laser2d.yaml` bounds `min_particles=500` and `max_particles=2000`. Seeded with a broad prior, the filter starts at the 2000-particle ceiling; once the first laser corrections concentrate the posterior the KLD bound (Fox) shrinks the set to the 500-particle floor, where it stays for the remainder of the closed-loop run across all three fixed seeds. Steady-state tracking therefore costs $4\times$ fewer likelihood evaluations per update than the fixed worst-case ceiling, with no loss of the accuracy shown in Fig. 2.

densities and constant biases the ESKF process model assumes (`fusion3d.yaml` defaults), plus fixed true biases the filter must estimate from zero; all noise derives from a single fixed seed (20240517). The IMU runs at 200 Hz, a simulated NDT source delivers 6-DoF pose updates at 10 Hz, and a simulated GNSS source delivers position updates at 1 Hz, with injected measurement noise matched to the filter’s R matrices. A 10 s window ($t \in [25, 35]$ s) withholds both aiding streams to expose IMU-only dead reckoning and recovery. Fig. 5 shows the position-error norm against the filter’s own 3σ position bound: windowed position RMSE is 0.016 m before, 0.125 m during, and 0.009 m after the dropout, with a peak error of 0.256 m at $t = 34.5$ s (end of the IMU-only blackout) recovering to 0.015 m within one aiding cycle by $t = 35$ s. The 3σ bound inflates from roughly 0.03 m under full aiding to a peak of 0.663 m at $t = 34.5$ s and stays above the true error throughout this run. A single run does not test whether the reported covariance is statistically consistent, so we make no consistency claim for it (Section VII).

Scope. This validation establishes that the estimators are individually correct, that all three sit behind one working ROS 2 contract, and that the assembled pipelines behave as intended in closed loop on synthetic data; it is deliberately not an accuracy benchmark. The planned quantitative field evaluation is discussed with the other roadmap items in Section VII.

VII. Limitations and Future Work

`prism_loc` is a systems paper that integrates established components, and its scope is deliberately bounded. We state the current boundaries plainly so that they read as a roadmap rather than as hidden assumptions.

Planar state in the MCL backends. Both particle-filter backends estimate a planar pose (x, y, θ) : the shared

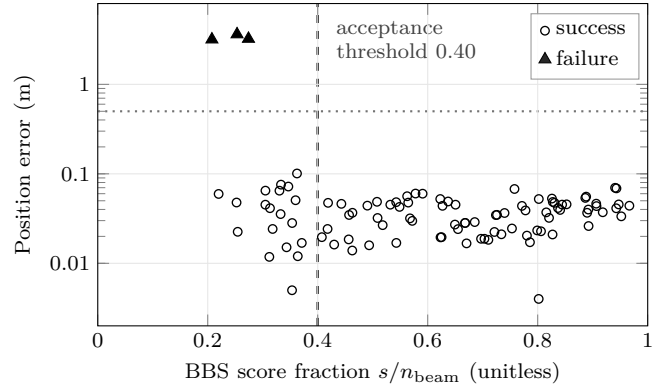


Fig. 4. Branch-and-bound (BBS) global relocalization for a kidnapped robot on a synthetic occupancy grid. From an asymmetric $6\text{ m} \times 6\text{ m}$ room (two interior wall stubs break the four-fold symmetry, the same world as the Bbs unit test), $N = 100$ true poses are drawn uniformly over free space with uniform yaw (fixed seed). For each pose a single 180-beam scan is produced by grid-march ray casting (`test_helpers.hpp`) and corrupted with synthetic clutter—each beam is, with a per-query probability drawn uniformly in $[0, 0.8]$, replaced by a spurious range—to span an easy-to-hard difficulty range. The real `prism_loc_core::BranchAndBoundMatcher` (`bbs.cpp`) then searches the whole map with the `laser2d.yaml` `bbs_*` defaults (± 10 m linear, $\pm \pi$ yaw at 1° steps, `max_depth=6`, `max_beams=120`). Each marker is one query: position error versus BBS score fraction (raw score over used beams); the dashed line is the acceptance threshold `bbs_min_score_fraction=0.40`. A query counts as a success when position error < 0.5 m and yaw error $< 10^\circ$. The score cleanly separates outcomes: all three failures score below 0.40 (with > 3 m error), and every match at or above the threshold is correct, so the threshold yields zero false accepts over the run (the 19 correct fixes it rejects are its conservative margin). Success rate is $97/100 = 97\%$; median full-map query latency is 37.7 ms (single-threaded, machine-dependent); median position error 0.037 m and median yaw error 0.48° .

core represents each particle as an $SE(2)$ element, and the `ndt3d` backend evaluates its per-particle likelihood against a full 3D Gaussian-voxel map from that planar pose at a fixed base height. This suits wheeled robots on locally flat floors but does not track roll, pitch, or vertical motion. Full $SE(3)$ state exists only in the `fusion3d` backend, whose 15-dimensional error state carries position, velocity, orientation, and IMU biases; existing particle-filter packages already estimate a full 3D pose against NDT or point-cloud maps [4], [5], [19]. Extending the MCL core to a 6-DoF particle state, or coupling the planar filter to an external attitude source, is future work.

2D-only global relocalization. The branch-and-bound search used for kidnapped-robot recovery (Section IV-B) operates on a rasterized 2D occupancy pyramid and returns an $SE(2)$ pose. There is no analogous 3D global localization for the `ndt3d` or `fusion3d` backends; those currently rely on an operator-supplied initial pose (via `/initialpose`) and local tracking. Relocalization runs only at startup or on a service call; unlike reliability-aware MCL [12], no loss detector triggers it. A 3D branch-and-bound front end such as 3D-BBS [28], or a learned place-recognition front end, is on the roadmap.

No online sensor time-offset estimation. The `fusion3d`

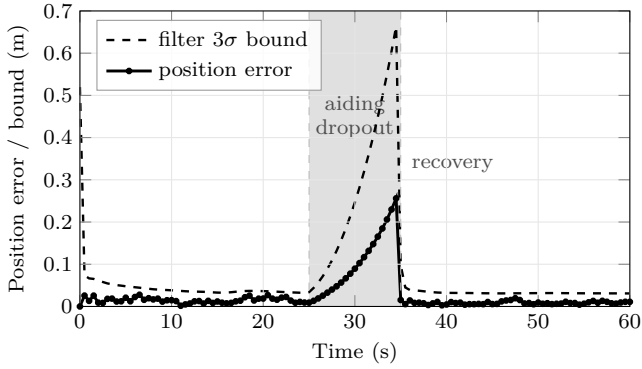


Fig. 5. Error-state Kalman filter (ESKF) fusion of a synthetic 200 Hz IMU with 10 Hz 6-DoF pose (NDT-like) and 1 Hz position (GNSS-like) updates over a smooth 60 s 3-D figure-eight trajectory. All data are synthetic: the IMU is generated analytically from the trajectory and corrupted with the white-noise densities and constant biases the ESKF process model assumes (fusion3d.yaml defaults), and every plotted value comes from the CSV written by experiments/eskf_fusion.cpp, which runs the real prism_loc_fusion::Eskf. Both aiding streams are withheld for $t \in [25, 35]$ s (shaded): the position error grows monotonically to 0.26 m on IMU dead-reckoning alone while the filter’s own 3σ bound inflates to 0.66 m, and both collapse within one aiding cycle once updates resume. Windowed RMSE (from the CSV): 0.016 m before, 0.125 m during, and 0.009 m after the dropout.

ESKF propagates on IMU samples and corrects with LiDAR-NDT and GNSS measurements at their stamped times, but it does not estimate the temporal offset between the LiDAR/GNSS clocks and the IMU clock, nor does it self-calibrate the LiDAR-IMU extrinsic; both are treated as known inputs. Jointly estimating time offset and extrinsics would improve robustness under poorly synchronized hardware.

Loosely-coupled LiDAR update and an unverified covariance. In fusion3d, PCL NDT registers each live cloud to the prior map to produce a 6-DoF pose measurement, which the ESKF then fuses as a loosely-coupled update. This keeps the ROS-free filter core independent of the registration engine, but it forgoes the accuracy of tightly-coupled formulations that fold raw scan residuals directly into the filter Jacobian, as in FAST-LIO [25]. The loose coupling is an architectural choice favoring modularity and testability over last-percent accuracy. The consistency of the covariance that the filter reports has not been established: the dropout experiment of Section VI-B is a single run and does not test it. We therefore regard fusion3d as experimental, and consumers should treat its published covariance with caution.

Single static map. All three backends assume one prior map of a static world, loaded once at startup; they neither update the map online nor reason about moving obstacles or long-term scene change. Map maintenance and change detection are out of scope for the present system.

No external baseline and no real data. The validation reported in Section VI is deterministic, ROS-free unit and integration testing plus continuous integration, comple-

mented by a seeded synthetic evaluation whose worlds and sensor-noise models are generated to match the filters’ assumptions. Every result in it is internal to prism_loc: we have not compared the trackers with nav2_amcl, Beluga [4], or MH-AMCL [13], or the relocalizer with hdl_global_localization [14], the recovery tooling of lidar_localization_ros2 [15], CBGL [30], or the method of Zhang et al. [31]. The study therefore establishes that each estimator core behaves as specified, including in closed loop; it does not establish accuracy on a physical robot or competitiveness with these systems. We have not yet measured trajectory error on physical or public datasets. The planned evaluation runs the three backends and at least nav2_amcl and one external global-localization method on public LiDAR sequences, including one with repeated structure, and reports absolute and relative trajectory error (ATE/RPE) per the TUM RGB-D conventions [35] on the per-backend map→odom estimates, together with relocalization success and false-accept rates. Until then we make no accuracy claims.

VIII. Conclusion

We presented prism_loc, a LiDAR localization stack whose contribution is a specific integration of established components rather than a new algorithm: three known estimator families — likelihood-field MCL, NDT-MCL [1], and an error-state Kalman filter that fuses LiDAR, IMU, and RTK-GNSS — are placed behind a single ROS 2 topic and TF contract, so that a 2D LiDAR, a 3D LiDAR, or a sensor-fused configuration is a deployment choice rather than a switch to a stack with a different interface. Each estimator is implemented as a pure C++ core with no dependence on the middleware runtime, making the filters unit-testable without ROS and letting the two MCL backends share one predict-weight-resample core that differs only in its measurement model, as in existing MCL libraries [4], and laser2d relocalizes deterministically by branch-and-bound [16], committing a match only when its score passes a threshold. We claim no new filter theory and no new software pattern; the contribution is this combination behind a familiar interface. An evaluation on real and public data against external baselines remains future work. prism_loc is released under the Apache-2.0 license at https://github.com/kjungmo/prism_loc.

Appendix Reproducibility

Every number and figure in Section VI-B regenerates from the repository alone; this appendix consolidates the recipe.

Environment. The experiment drivers and the core test suites need only a C++17 compiler and the Eigen 3 headers (Debian/Ubuntu packages g++ and libeigen3-dev); the unit tests of Section VI-A additionally use GoogleTest (libgtest-dev). ROS 2 Humble is required only for the two

TABLE III

Experiment drivers, fixed seeds, and committed artifacts. Drivers live under experiments/; CSVs under docs/paper/data/.

Driver (.cpp)	Seed(s)	Figure(s)	CSV(s)
mcl_tracking	101, 202, 303	2, 3	mcl_tracking, mcl_particles
bbs_relocalization	20240607	4	bbs_relocalization
eskf_fusion	20240517	5	eskf_track

node packages; continuous integration builds those in a ros:humble container (.github/workflows/ci.yml).

Building and running the drivers. The three drivers compile the estimator cores directly from source—no CMake, no ROS. From the repository root:

```
g++ -O2 -std=c++17 -I prism_loc_core/include \
-I prism_loc_core/test \
-I /usr/include/eigen3 \
prism_loc_core/src/*.cpp \
experiments/mcl_tracking.cpp -o /tmp/exp_mcl
/tmp/exp_mcl
```

```
g++ -O2 -std=c++17 -I prism_loc_core/include \
-I prism_loc_core/test \
-I /usr/include/eigen3 \
prism_loc_core/src/*.cpp \
experiments/bbs_relocalization.cpp \
-o /tmp/exp_bbs
/tmp/exp_bbs
```

```
g++ -O2 -std=c++17 \
-I prism_loc_fusion/include \
-I /usr/include/eigen3 \
prism_loc_fusion/src/*.cpp \
experiments/eskf_fusion.cpp -o /tmp/exp_eskf
/tmp/exp_eskf > docs/paper/data/eskf_track.csv
```

All randomness derives from fixed compile-time seeds (Table III), so every CSV column is bit-for-bit reproducible, with the single exception of the wall-clock time_ms latency column of the BBS experiment, which is machine-dependent.

Parameters. The drivers mirror the shipped defaults. Every ROS 2 parameter and its in-code default is documented in PARAMS.md at the repository root; the deployed overrides are the YAML files prism_loc/params/laser2d.yaml, prism_loc/params/ndt3d.yaml, and prism_loc_fusion_ros/params/fusion3d.yaml.

Number tracing. The committed CSVs under docs/paper/data/ are the artifacts of record. The guard script docs/paper/check_numbers.py recomputes every headline statistic quoted in Section VI-B from those CSVs and exits nonzero if any quoted number no longer matches, so the paper’s statistics cannot silently drift from the data.

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